INCH-POUND

MIL-DTL-7034/4B
10 March 2022
SUPERSEDING
MIL-DTL-7034/4A
w/Amendment 2
9 April 2012

DETAIL SPECIFICATION SHEET

PULLEY, CONTROL, ANTI-FRICTION BEARING, AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet, MIL-DTL-7034, and QPL-7034.

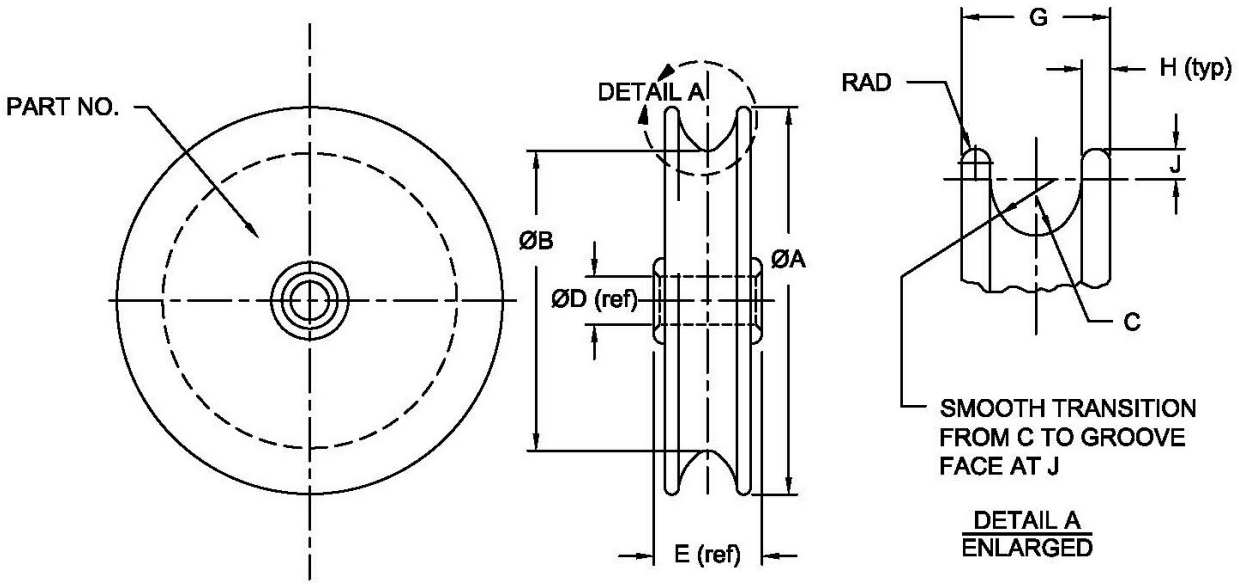


FIGURE 1. Configuration and dimensions.

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TABLE I. Dimensions and interface characteristics.

Dash no. 1/ Nonmetallic sheave	Cable size (ref) (inches)	A Outside dia +.000 -.010	B Groove dia +.000 -.010	C Groove rad	D Bore dia (ref) 2/	E Width I.R. (ref) 2/	G Width sheave
-1B	1/16 3/32	1.250	.972	.055 +.000 -.003	.19	.30	.250 +.000 -.012
-2B		2.500	2.222				
-3B	1/8 5/32 3/16	2.000	1.510	.112 +.000 -.003	.25	.48	.422 +.000 -.012
-4B		3.500	3.010				
-5B	3/16 7/32 1/4	5.000	4.374	.145 +.000 -.005	.38	.62	.507 +.000 -.014
-6B		6.000	5.374				

TABLE I. Dimensions and interface characteristics - Continued.

Dash no. 1/ Nonmetallic sheave	H Flange width	J ±0.005	Pulley assembly weight max (lbs)	Pulley Strength Allowable limit load on pulley (lbs)	SAE bearing
-1B	0.060 ±0.005	0.040	0.026	300	MS21443-3A 3/
-2B			0.075	500	MS21443-3B 3/
-3B	0.086 ±0.006	0.086	0.090	600	MS21443-4C 3/
-4B			0.200	1,200	
-5B	0.092 ±0.007	0.092	0.500	3,000	MS27640-6 4/
-6B			0.660	4,000	

1/ See note 3 for example of complete part number.
2/ Bearing bore diameter and bearing width are controlled by the applicable SAE specification. See notes 3/ and 4/.
3/ Bearing per SAE AS21443
4/ Bearing per SAE AS27640

REQUIREMENTS:

1. Material: Material shall be in accordance with MIL-DTL-7034.

2. Finish: Finish shall be in accordance with MIL-DTL-7034.

3. Surface roughness: Surface roughness shall be in accordance with MIL-DTL-7034.

4. Hub may be added to pulley sheave where width of bearing outer race is greater than the maximum width of the pulley sheave.

NOTES:

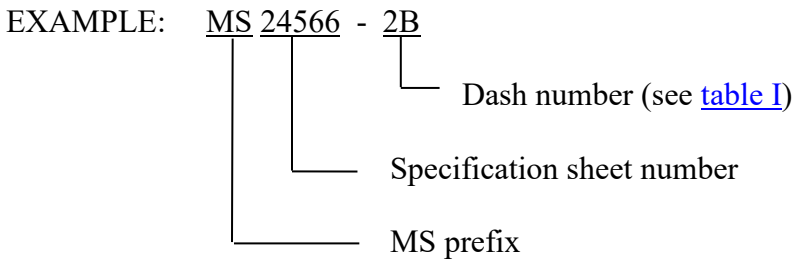
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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1. Dimensions are in inches. Unless otherwise specified, decimal tolerances: +0.000, -0.010. End faces of all bearing hubs shall be flat and square with bore within ±1°.

2. -1B and -3B pulleys should not be installed on aircraft controls that bend the cable more than 15 degrees from a straight line.

3. The part or identifying number (PIN) to be used for pulleys acquired to this specification is created as shown below. In the transition of MS24566 to detail specification sheet MIL-DTL-7034/4, the MS24566 part numbers have been retained as follows:



MS24566-2B Indicates: Size 2 nonmetallic sheave and fixed ball bearing with contact seal.

All dash numbers in table I are identical to and interchangeable with like dash numbers of MS24566.

4. The replacement sequence for this document is AN210 replaced by MS24566 replaced by MIL-DTL-7034/4.

5. For design feature purposes, this specification sheet takes precedence over the acquisition document referenced herein.

CHANGES FROM PREVIOUS ISSUE. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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(Project 1640-2022-007)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil/.

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